Implementation of Fuzzy Tsukamoto Algorithm on Smart Node Sensors for Air Quality Monitoring

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***Abstract*—In this modern era, there are more and more industrial centers in urban areas as well as the development of various transportation. This is causes air pollution that can affect human, animal, plant life. The initial study of the application of the Fuzzy Tsukamoto algorithm on smart sensor node system for air quality monitoring described in this paper. The air parameters used are CO, CO2, and CH4. The data used were obtained from previous studies. This data is used to test the accuracy of the fuzzy model using two rule bases and two types of fuzzy. The results show that the accuracy value of Fuzzy Tsukamoto using the second rule base is better, namely 92.4 percent.The classification results are in the form of warning alerts that are combined in one frame with other data on the raspberry pi. The data are sent to the PC server using LoRa transceivers. From indoor and outdoor testing, the results show that the system can work well. In indoor conditions, the results of the air classification are Good with CO concentrations of not more than 0.9 ppm, CO2 concentrations of 293-402 ppm and CH4 concentrations of 0.5 ppm. In outdoor conditions, the results obtained are Moderate air classification with CO concentrations of not more than 1.0-1.1 ppm, CO2 concentrations of 707-915 ppm, and CH4 concentrations of 1.6-2.3 ppm.**

***Keyword*—Fuzzy Tsukamoto, Air Quality, Sensor Node, LoRa**

I. Introduction

Air is one of the primary needs for human, animal, and plant life. Therefore, the air must be maintained so as not to be polluted and caused bad effects. However, along with the development of industry in big cities and more and more modes of transportation, the risk of air pollution gets bigger. This is a serious problem because it can affect the environment, human health and even cause death. According to the World Health Organization (WHO) that nearly 2.4 million people die every year due to air pollution[1]. Some gases such as CO, CO2, and CH4 can cause death if inhaled continuously by humans. Carbon Monoxide (CO) and carbon dioxide (CO2) are producing from exhaust gases from vehicles or factories[2]. Based on data from the World Wide Fund for Nature International, the amount of carbon dioxide emissions increased 12 times greater than in 1900[3]. In addition, the use of fuel also produces other harmful gases, namely methane gas (CH4). Based on Indonesia's Environmental Statistics Data in 2017, which is a quote from kompas.com, it is mention that big cities in Indonesia produce around 962 m3 of organic waste per day and produce 94.64 methane gas[4].

A monitoring device model using PM 2.5, NO2, and CO sensors is proposing. The device is using in indoor conditions by using a microcontroller. From the three sensors, the output is in the form of a buzzer sound or fan flame according to a predetermined value range[5]. In general, the sensor model used for air monitoring does not include all types of sensors that are urgent to use, such as the MQ-7 and MQ-4 sensors. These two sensors are very important to detect the content of CO2 gas and CH4 gas, both of which are very dangerous for humans and the environment. If these two gases exceed the threshold it can cause death.

Prototype Of The Measuring Device Carbon Monoxide (CO) in Smoking Room Using Fuzzy Tsukamoto has been proposed. This tool is used to control the air quality in a smoking room or indoor conditions. In this case, the carbon monoxide sensor sends data, and then it is processed using Fuzzy Tsukamoto. The output results obtained will be displayed on the LCD. When data processing using the Fuzzy Tsukamoto algorithm exceeds the predetermined value limit, it will activate the buzzer and fan to remove carbon monoxide levels in the room [6]. In general, there is no data processing presented that produces an intelligent sensor node that provides decision making from an outdoor environmental condition or provides a warning. In addition, there is no direct data transmission to the gateway using LoRa transmission media.

In this paper, an air pollution monitoring system based on Wireless Sensor Network will be presented. The system utilizes transmission technology in the form of LoRa and is also equipped with the Fuzzy Tsukamoto algorithm to obtain classification results from air pollution sensor values. This device is applied outdoor using the MQ-7 sensor to detect carbon monoxide gas, MQ-135 sensor to detect carbon dioxide gas, and MQ-4 sensor to detect methane gas. The sensor provides data to the Arduino which is sent to the raspberry pi for the classification process using Fuzzy Tsukamoto. In each sensor, there are four members representing Good, Moderate, Unhealthy, and Dangerous. Determination of each sensor member using AQI according to the range on each

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sensor. Classification of air quality in the node will provide a status according to the level of pollutants detected by the sensor. The resulting air quality output consists of five classifications, namely Good, Moderate, Unhealthy, Very Unhealthy, or Dangerous. The results of the classification are in the form of warning alerts that are combined in one frame with ID, time, and sensor data by a raspberry pi. The data are sending to a PC server by utilizing a medium-distance communication system using a LoRa transceiver.

For further discussion in this paper consists of four parts. The beginning contains an introduction that explains the background and related research. The second part is a system overview explaining the overall system design and the stages of the Fuzzy Tsukamoto algorithm on the sensor node. The third part is the result which contains system testing and analysis. And at the end, the conclusions and plans for the next research will be presented.

II. System Overview

A. Overall System Design

The smart node sensor in this study is shown by the block diagram in Fig 1, which explains the overall system design from sending to receiving data. However, this research will focus on smart node sensors for monitoring air quality using the Fuzzy Tsukamoto algorithm. The sensors used are the MQ-7 sensor to measure Carbon Monoxide gas levels in the air, MQ-135 which is used to measure Carbon Dioxide gas levels in the air, and MQ-4 which is used to measure Methane gas levels in the air. The three sensors are connected to a microcontroller or Arduino to process the values obtained. Sensor value from the Arduino will be communicated serial between the Arduino and the raspberry pi. Next, add ID, date, and time. ID is using to find out what sensor data is sending to the gateway, and will be stored in a file.txt. Furthermore, data processing is carried out using the Fuzzy Tsukamoto algorithm. There are five classifications of output used, namely Good, Moderate, Unhealthy, Very Unhealthy, or Dangerous. Sending data to the gateway is by utilizing the remote WSN technology, namely LoRa.

B. Fuzzy Tsukamoto Algorithm on Sensor Node

In processing sensor data to obtain a classification of air quality, the Fuzzy Tsukamoto algorithm is using. In the Fuzzy Tsukamoto algorithm, there are several stages to get the Classification, namely Fuzzification, making Rule-Based, Inference Machine, Defuzzification, and Classification will be obtained from the calculated sensor values. The flowchart of the stages of data processing using the Fuzzy Tsukamoto algorithm is showing in Fig 2. For the classification process itself, there are five classifications, namely Good (G), Moderate (M), Unhealthy (U), Very Unhealthy (VU), and Dangerous (D).

Fuzzification

Processing the sensor data using the Fuzzy Tsukamoto algorithm begins with the determination of the fuzzy sets (Fuzzification). The sets used to consist of two, namely the input set and the output set. The input sets consist of fuzzy input sets for the sensor variables for Carbon Monoxide, Carbon Dioxide, and Methane. The fuzzy output set is the air quality variable. The membership functions used in this research are descending linear curves, triangular curves, and ascending linear curves. Where for the explanation of each curve is as follows.

For a descending linear curve, when the set decreases from one degree of membership, it moves to the right towards a lower membership towards zero, as shown in Fig 3.

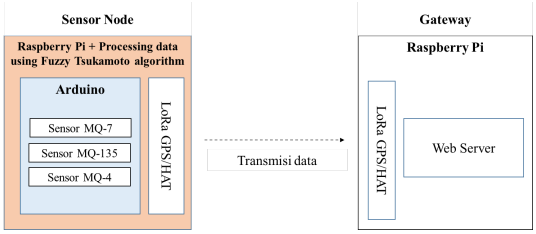

Fig. 1. System Design Node Sensor.

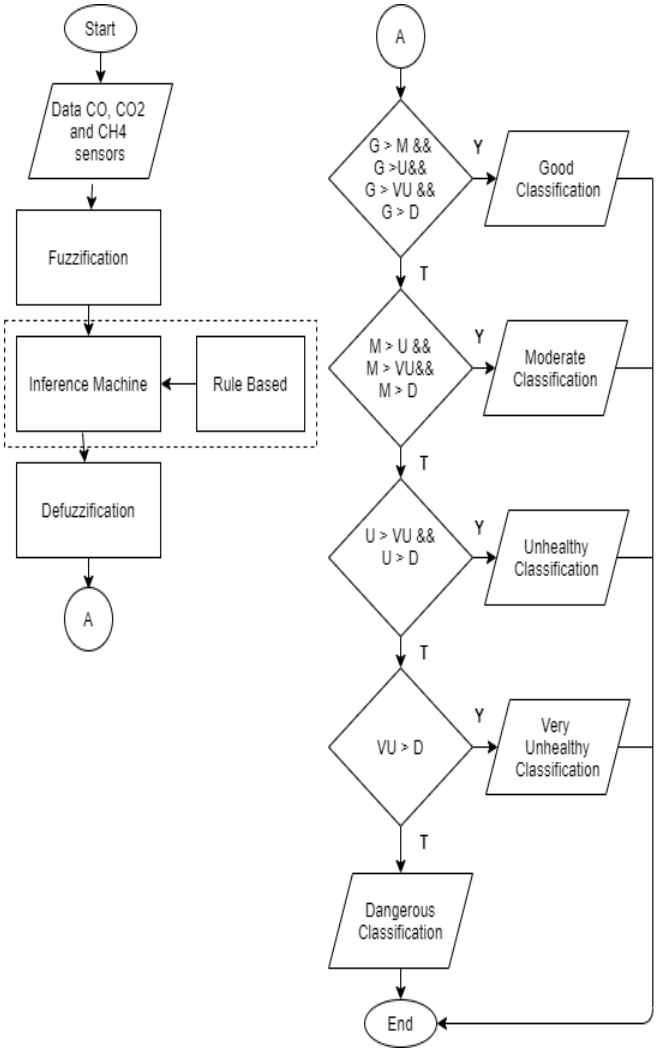

Fig. 2. Fuzzy Tsukamoto Algorithm Flowchart.

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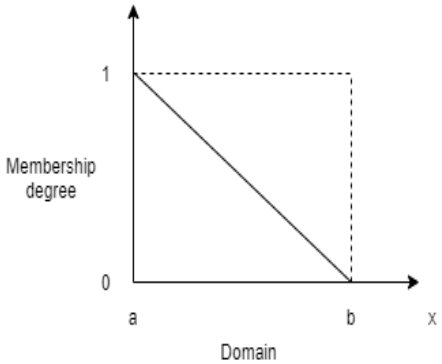

Fig. 3. Descending Linier Curves.

To get the membership value when using a descending linear curve by using the (1). For variable a = the smallest domain value when membership degree is smallest and variable b = the largest degree of membership in the domain.

$$\mu(x) = \begin{cases} 1, & ; x \leq a \\ \frac{b-x}{b-a}, & ; a \leq x \leq b \\ 0, & ; x \geq b \end{cases} \tag{1}$$

The representation of a triangular curve is a combination of a descending linear representation and an ascending linear curve, as shown in Fig 4. To get the value of the degree of membership when using a triangular curve by using the (2). For variable a = the smallest domain value when membership degree is small, variable b = the largest degree of membership in the domain and variable c = the largest domain value when the membership degree is small.

$$\mu(x) = \begin{cases} 0, & ; x \leq a \ atau \ x \geq c \\ \frac{x-a}{b-a}, & ; a \leq x \leq b \\ \frac{c-x}{c-b}, & ; b \leq x \leq c \end{cases} \tag{2}$$

An ascending linear curve means the set has increased from zero membership degrees moving to the right towards higher membership degrees towards one, as shown in Fig 5.

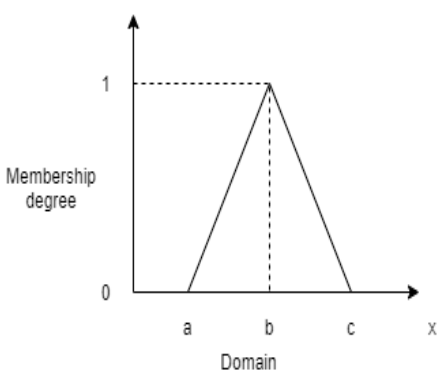

Fig. 4. Triangular Curves.

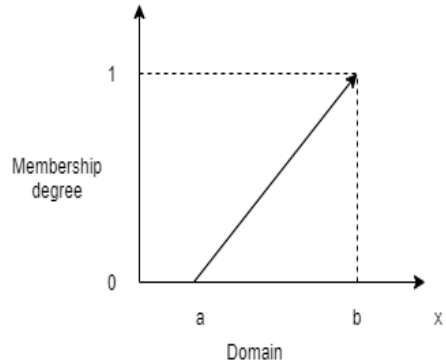

Fig. 5. Ascending Linear Curves.

To get the value of the degree of membership when using an ascending linear curve by the using (3). For variable a = the smallest domain value when membership degree is small and variable b = the largest degree of membership in the domain.

$$\mu(x) = \begin{cases} 0, & ; x \leq a \\ \frac{x-a}{b-a}, & ; a \leq x \leq b \\ 1, & ; x \geq b \end{cases} \tag{3}$$

The fuzzy set for the carbon monoxide sensor variable consists of a membership function of a descending linear curve, a triangular curve, and an ascending linear curve. For the domain value, which is classification as a descending linear curve, namely 0.8-2 ppm, the domain value for the triangular curve is from 0.8-8.5 ppm and 7-10 ppm, while for the linear curve it is increasing from 9-15 ppm, As shown in Fig 6.

The fuzzy set for the carbon dioxide sensor variable consists of a membership function of a descending linear curve, a triangular curve, and an ascending linear curve. For the domain value belonging to a descending linear curve, namely 400-606 ppm, the domain value for a triangular curve is from 580-1020 ppm and 800-1520 ppm, while for an increasing linear curve, it is from 1480-1800 ppm, As shown in Fig 7.

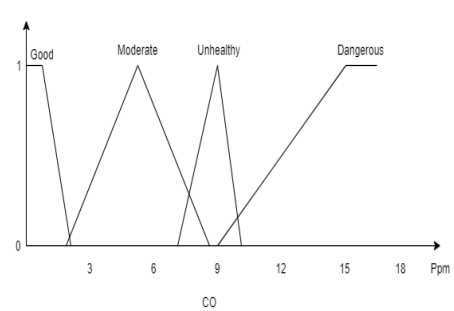

Fig. 6. CO Fuzzy Graph.

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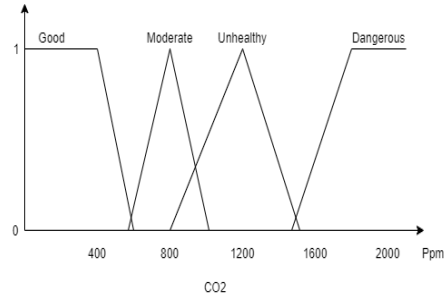

Fig. 7. CO2 Fuzzy Graph.

The fuzzy set for the methane sensor variable consists of a membership function of a descending linear curve, a triangular curve, and an ascending linear curve. For the domain value, which is classification as a descending linear curve, namely 0.5-2 ppm, the domain value for the triangular curve is from 1-5 ppm and 4-8 ppm, while for the linear curve it is increasing from 7-10 ppm, As shown in Fig 8.

The fuzzy output set for the air quality variable consists of one descending linear curve, three triangular curves and one ascending linear curve. Members of the fuzzy output consist of Good (G), Medium (M), unhealthy (U), very unhealthy (VU) and dangerous (D). The range of domain values for linier curves decreased by 30-50 ppm, for triangular curves they were 30-100 ppm, 80-200 ppm, and 180-300 ppm. while for ascending linier curves by 300-500 ppm.Air quality fuzzy set as shown in Fig 9.

Rule Based and Inference Machine

There are 64 rules used in this research. As for the inference machine stage, the formula for the membership sets is using for the variable carbon monoxide sensor, carbon dioxide sensor, and methane sensor. The calculation of each member of the sets is then entered into each rule-based and can be used to get the value of a-predicate by using the MIN method. And from the MIN value obtained, it is using to find the z value using the formula from the membership function for the air quality variable.

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To get the centroid value, you can use the (4). After the Z value has been obtained, then it is used to get the classification.

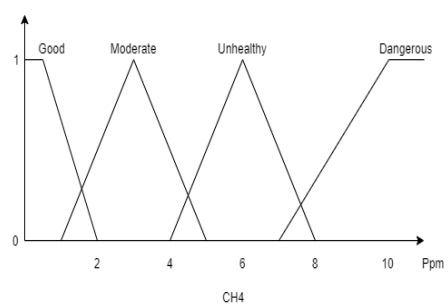

Fig. 8. CH4 Fuzzy Graph.

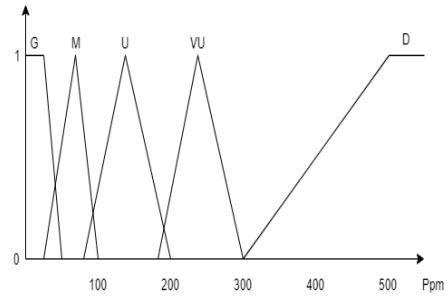

Fig. 9. Air Quality Fuzzy Graph.

From the Z value, it will be used to obtain a classification by calculating using the Membership Function for the air quality variable. From the five members in the set, the value is sought using the MAX method or the highest value between Good, Moderate, Unhealthy, Very Unhealthy, and Dangerous. The result of the calculation of the MAX method is the result of the classification obtained.

$$z = \frac{(\alpha Predikat2 * z1) + ... + (\alpha Predikatn * zn)}{\alpha Predikat1 + ... + \alpha Predikatn} \quad (4)$$

III. RESULT

A. Fuzzy Algorithm Accuracy Test

Tests were carried out in this study to determine the accuracy, before the algorithm to be used was implemented on the system. Testing the accuracy of this algorithm uses two types of fuzzy algorithms, namely Fuzzy Sugeno and Fuzzy Tsukamoto. Furthermore, it employs two types of rule-based systems, each of which is based on the Air Quality Index (AQI), which is divided into five categories: Good (G), Moderate (M), Unhealthy (U), Very Unhealthy (VU), and Dangerous (D). For the number of membership function outputs on the rule-based used, the details are as in Table 1.

TABLE I. RULES BASED ON EACH CIASSIFICATION.

Rule Based	G	M	U	VU	D
1	4	12	24	17	7
2	5	13	19	19	8

Table 2 shows the accuracy values obtained from the two types of fuzzy algorithm and the two rule-based methods used. Fuzzy Tsukamoto test using the second rule base is better accuracy, namely 92.4 percent. So that in the air quality monitoring system to produce air quality level classifications, the Fuzzy Tsukamoto algorithm is used.

TABLE II. FUZZY ALGORITHM ACCURACY RESULTS.

Rule Based	Fuzzy Sugeno	Fuzzy Tsukamoto
1	53.5	74
2	83.2	92.4

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B. Decision-Making

The data that has been received by the raspberry will be added to the classification results using the Fuzzy Tsukamoto algorithm. The classification process is carried out on the raspberry pi device by breaking the data first, then only sensor data is taken for classification using the Fuzzy Tsukamoto algorithm. There are five types of classification results, namely Good, Moderate, Unhealthy, Very Unhealthy and Dangerous. So that the format of the data frame that is sent is not too long for the data classification results, a code that states air quality is used. Where if the classification results produce a code of 0, then the air quality is ”Good”, when the classification results produce code 1, then the air quality is ”Moderate”, when the classification results produce code 2, then the air quality is ”Unhealthy”, when the classification results produce code 3, then the air quality ”Very Unhealthy” while if the classification results produce code 4, then the air quality is ”Dangerous”. The mechanism for inserting classified data are breaking all the data into an array. After all the data entered in accordance with the order of the array, the data are rewritten into a file.txt, so that the format of the data frame sent to the receiver is as shown in Fig 10.

Result Classification	,	ID Sensor	,	date-month-year	,	Time	,	Sensors data

Fig. 10. Format Frame Data.

C. Testing System

The system was tested twice, once in an indoor environment and once outdoor in a polluted environment. This is done to see if the algorithms and sensors can accurately classify air quality in both indoor and outdoor settings. The sensor node sends data to the receiver for system testing, as shown in Fig 11. Every 30 seconds, the sensor node sends data.

Indoor testing was carried out in the postgraduate laboratory on the 10th floor of the Surabaya State Electronics Polytechnic where the transmitter and receiver were in the same room. This test is performed to measure the indoor air concentration and by sending it directly to the gateway, the sensor node side as shown in Fig 12. From the test results, it is known that the air quality conditions in the postgraduate lab on the 10th floor are for carbon monoxide parameters of 0.9 ppm, carbon dioxide levels range from 293-402 ppm, while for methane levels, they range from 0.4-0.5 ppm. From the three conditions, the classification sensor value obtained for air quality in the lab is 0 which means the air quality is ”Good”. For the data sent by the sensor node is received by the receiver as shown in Fig 13.

Outdoor testing was carried out on the Semolowaru highway. Where traffic conditions are very congested. This test is carried out to determine the air quality on the Semolowaru highway, the sensor node side as shown in Fig 14. From the test results on the Semolowaru highway, it is known that the levels of air parameters used are higher than the levels of indoor air parameters. The carbon monoxide content was between 1.0-1.1 ppm, the carbon dioxide content was between 707-915 ppm, while the methane content was 1.6-2.3 ppm. From the levels of each gas detected by the sensor, the results of the air quality classification on the Semolowaru highway are ”Moderate”. This is due to the large number of vehicles passing on the Semolowaru highway so that the level of air quality becomes moderate. For the data sent by the sensor node is receiv well by the receiver as shown in Fig 15.

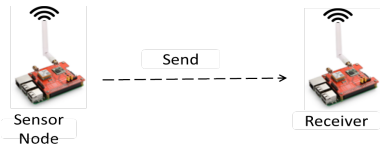

Fig. 11. Test Scenario.

```
send: 0,a,2021-06-15,19:48:00,00.9,389.0,00.5
send: 0,a,2021-06-15,19:48:30,00.9,397.1,00.5
send: 0,a,2021-06-15,19:49:00,00.9,383.6,00.5
send: 0,a,2021-06-15,19:49:30,00.9,397.1,00.5
send: 0,a,2021-06-15,19:50:00,00.9,402.6,00.5
send: 0,a,2021-06-15,19:50:30,00.9,328.0,00.4
send: 0,a,2021-06-15,19:51:00,00.9,309.2,00.4
send: 0,a,2021-06-15,19:51:30,00.9,304.7,00.4
send: 0,a,2021-06-15,19:52:00,00.9,332.8,00.4
send: 0,a,2021-06-15,19:52:30,00.9,293.4,00.4
```

Fig. 12. Transmitter Side.

```
Packet RSSI: -94, RSSI: -118, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:48:00,00.9,389.0,00.5
Packet RSSI: -97, RSSI: -120, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:48:30,00.9,397.1,00.5
Packet RSSI: -97, RSSI: -116, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:49:00,00.9,383.6,00.5
Packet RSSI: -97, RSSI: -118, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:49:30,00.9,397.1,00.5
Packet RSSI: -99, RSSI: -118, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:50:00,00.9,402.6,00.5
Packet RSSI: -97, RSSI: -118, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:50:30,00.9,328.0,00.4
Packet RSSI: -96, RSSI: -110, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:51:00,00.9,309.2,00.4
Packet RSSI: -96, RSSI: -120, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:51:30,00.9,304.7,00.4
Packet RSSI: -98, RSSI: -118, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:52:00,00.9,332.8,00.4
Packet RSSI: -97, RSSI: -118, SNR: 9, Length: 39
Payload: 0,a,2021-06-15,19:52:30,00.9,293.4,00.4
```

Fig. 13. Receiver Side.

```
send: 1,a,2021-06-26,15:59:00,01.0,707.0,01.6
send: 1,a,2021-06-26,15:59:30,01.0,702.7,01.6
send: 1,a,2021-06-26,16:00:00,01.0,737.6,01.7
send: 1,a,2021-06-26,16:00:30,01.0,769.2,01.8
send: 1,a,2021-06-26,16:01:00,01.0,778.4,01.8
send: 1,a,2021-06-26,16:01:30,01.0,797.0,01.9
send: 1,a,2021-06-26,16:02:00,01.0,801.7,01.8
send: 1,a,2021-06-26,16:02:30,01.1,859.7,02.0
send: 1,a,2021-06-26,16:03:00,01.1,915.4,02.2
send: 1,a,2021-06-26,16:03:30,01.1,952.2,02.3
```

Fig. 14. Transmitter Side.

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```
Packet RSSI: -45, RSSI: -103, SNR: 9, Length: 39
Payload: 1,a,2021-06-26,15:59:00,01.0,707.0,01.6
Packet RSSI: -46, RSSI: -109, SNR: 8, Length: 39
Payload: 1,a,2021-06-26,15:59:30,01.0,702.7,01.6
Packet RSSI: -46, RSSI: -102, SNR: 9, Length: 39
Payload: 1,a,2021-06-26,16:00:00,01.0,737.6,01.7
Packet RSSI: -44, RSSI: -102, SNR: 9, Length: 39
Payload: 1,a,2021-06-26,16:00:30,01.0,769.2,01.8
Packet RSSI: -43, RSSI: -102, SNR: 4, Length: 39
Payload: 1,`,R028-06-26-1>:01:00,01.0,778.4,01.8
Packet RSSI: -45, RSSI: -101, SNR: 9, Length: 39
Payload: 1,a,2021-06-26,16:01:30,01.0,797.0,01.9
Packet RSSI: -46, RSSI: -103, SNR: 9, Length: 39
Payload: 1,a,2021-06-26,16:02:00,01.0,801.7,01.8
Packet RSSI: -47, RSSI: -102, SNR: 9, Length: 39
Payload: 1,a,2021-06-26,16:02:30,01.1,859.7,02.0
Packet RSSI: -48, RSSI: -102, SNR: 8, Length: 39
Payload: 1,a,2021-06-26,16:03:00,01.1,915.4,02.2
Packet RSSI: -45, RSSI: -104, SNR: 9, Length: 39
Payload: 1,a,2021-06-26,16:03:30,01.1,952.2,02.3
```

Fig. 15. Receiver Side.

IV. Conclusion

In this research, the Fuzzy Tsukamoto algorithm has been implemented on the sensor node for monitoring air quality. The accuracy results obtained in the Tsukamoto fuzzy learning process are 92.4 percent. The results of decision making using Fuzzy Tsukamoto have shown results that following the system design. The classification results are displayed in a data frame format for early warning about air quality in real-time. Tests have also been carried out in indoor and outdoor conditions where the Tsukamoto fuzzy algorithm and sensors work well, in indoor conditions the air quality results are ”Good” while in outdoor conditions the air quality results are ”Moderate”.

In further research, the following activities will be developed: data processing using an algorithm that has a higher accuracy value so that the classification results are more accurate. The addition of the number of nodes spread over several points to be tested can provide a representative picture of environmental conditions.

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